FPV Reconstruction - Validation Summary

Final relative-reconstruction baseline | Google Colab | VGGT-Omega | True FP32

Baseline status: ready for review Four deliberately different FPV videos completed with auditable outputs. The notebook preserves warnings and unavailable intervals instead of hiding or inventing motion.

Representative validation runs

Video	Purpose	Trusted poses	Geometry	Result	Main observation
Adaissah	Clear reference	25 / 25	1 component	**PASS***	Coherent forward corridor; corrected true-scale path.
Namer / Khiam	Fast bank and descent	27 / 27	1 component	**PASS***	Motion warnings correspond to visible hard movement.
Kfar Tebnit	Difficult scene transition	40 / 40	1 component	**PASS***	Large warned jump is shown as a break, not connected.
Majdal Zoun	Different terrain and object	27 / 28	1 component	**PASS***	One final weak pose masked; remaining path is coherent.

What was verified

- Video download, cleanup, and frame selection
- VGGT-Omega reconstruction on a Tesla T4
- GLB, PNG, XYZ CSV, audit files, and ZIP outputs
- Trusted-pose, motion, continuity, and GPU diagnostics

Trajectory safeguards

- True-proportion 3D plus equal-scale top and side views
- No smoothing of direct VGGT poses
- No line through removed intervals or warned jumps
- Raw coordinates remain available for audit

Current deliverable

One clean Colab notebook accepts a direct video URL, runs the complete pipeline, saves results to Drive, and presents the important outputs together at the end.

Important boundary

XYZ and derived speed are relative model units. They are not metres, physical speed, latitude, longitude, or altitude. Metric scale and geographic alignment are separate future stages.

* PASS with visible warnings retained for review Validated through 31 August 2026